

# CS611 FINAL PROJECT  
## LEGENDS OF VALOR

---

Name: Serena Noboudem  
Email: [serenanm@bu.edu](mailto:serenanm@bu.edu)  
ID: U63767551

Name: Chris Mary Benson  
Email: [chris27@bu.edu](mailto:chris27@bu.edu)  
ID: U56085268

Name: Ziyang Wang  
Email: [zywang1@bu.edu](mailto:zywang1@bu.edu)  
ID: U12285471

**Github Link:** <https://github.com/N-serena/LegendsOfValors>

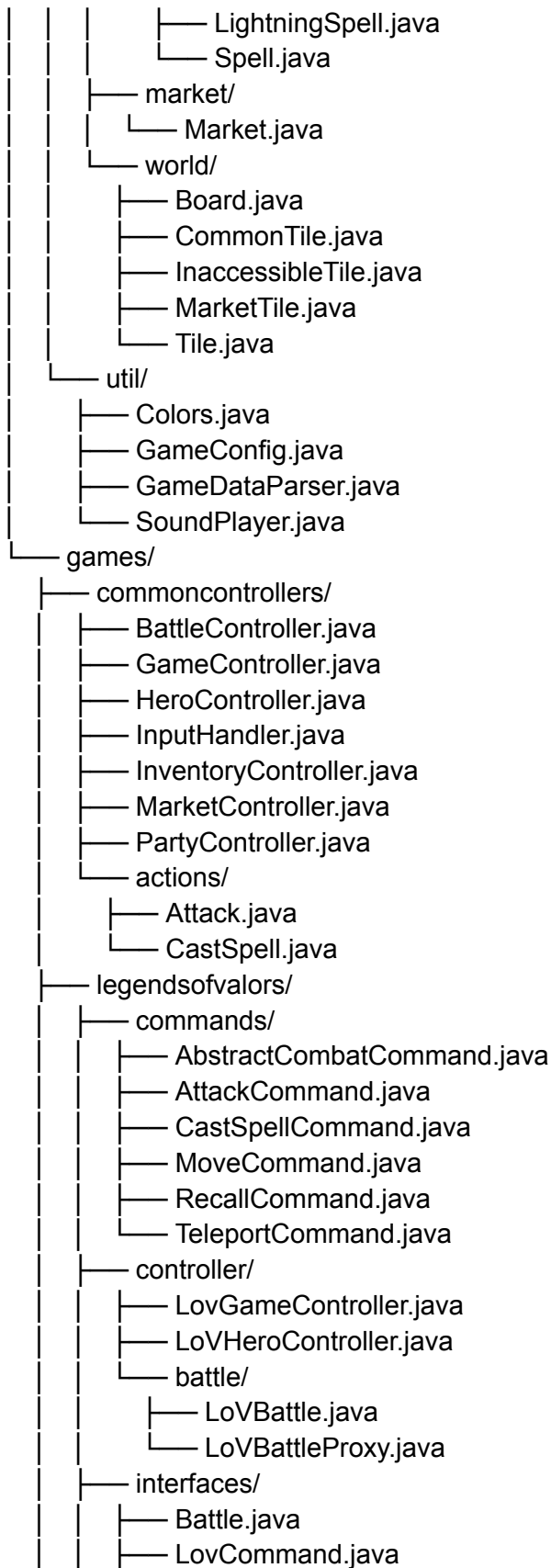
## Files

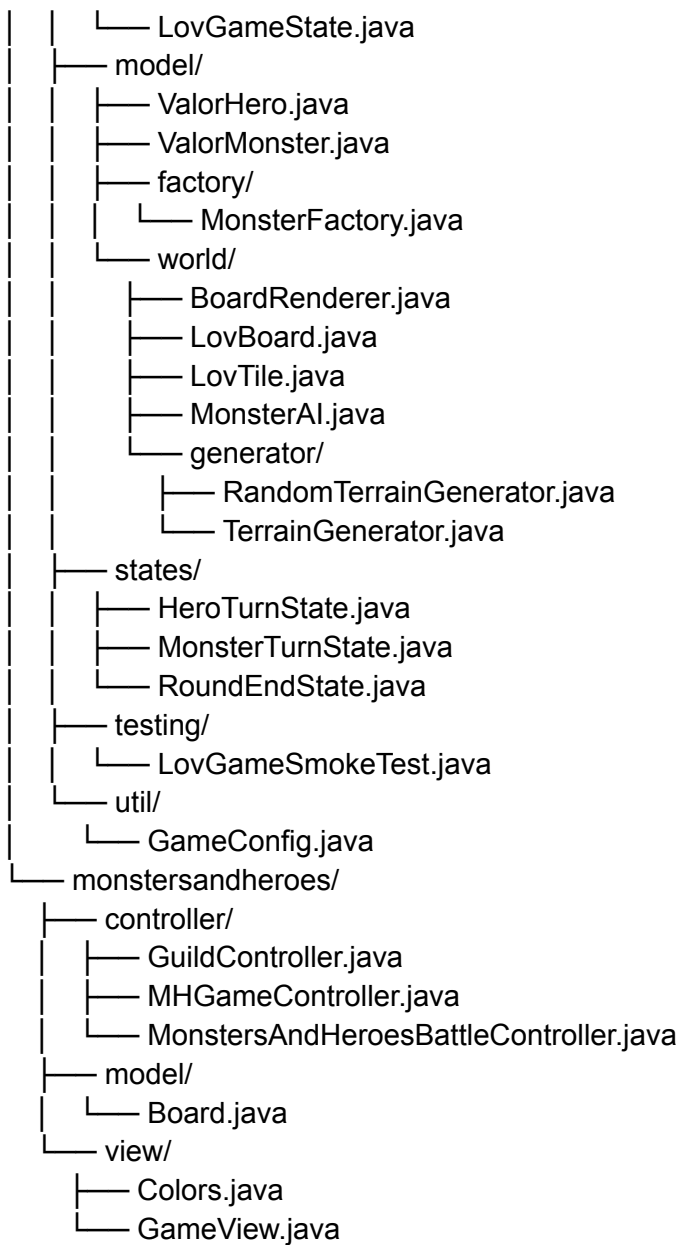
---

LegendsOfValors

- ├── ReadMe\_FinalProject.md
- ├── out/
- ├── LegendsOfValors
  - ├── .idea/
    - ├── .gitignore
    - ├── aws.xml
    - ├── copilot.data.migration.agent.xml
    - ├── copilot.data.migration.ask.xml
    - ├── copilot.data.migration.ask2agent.xml
    - ├── copilot.data.migration.edit.xml
    - ├── misc.xml
    - ├── modules.xml
    - ├── uiDesigner.xml
    - └── vcs.xml
  - ├── HeroesAndMonsters.iml
  - ├── ReadMe\_FinalProject.md
  - ├── data\_files/
    - ├── Armory.txt
    - ├── Dragons.txt
    - ├── Exoskeletons.txt
    - ├── FireSpells.txt
    - ├── IceSpells.txt
    - ├── Initial.txt
    - └── LightningSpells.txt

```
|
|
|— Paladins.txt
|— Potions.txt
|— Sorcerers.txt
|— Spirits.txt
|— Warriors.txt
|— Weaponry.txt
|
|— out/
|— out8/
|— runinstruction.png
|— soundfiles/
|   |— themes.wav
|
|— src/
|   |— GameLauncher.java
|   |— Main.java
|   |— core/
|       |— interfaces/
|           |— Board.java
|           |— FightStrategy.java
|           |— GameEngine.java
|           |— HeroEventNotifier.java
|           |— HeroObserver.java
|       |— model/
|           |— GameDatabase.java
|           |— Party.java
|           |— entity/
|               |— Dragon.java
|               |— Exoskeleton.java
|               |— Hero.java
|               |— LivingEntity.java
|               |— Monster.java
|               |— Paladin.java
|               |— Sorcerer.java
|               |— Spirit.java
|               |— Warrior.java
|               |— decorator/
|                   |— HeroDecorator.java
|           |— item/
|               |— Armor.java
|               |— Item.java
|               |— Potion.java
|               |— Weapon.java
|               |— spell/
|                   |— FireSpell.java
|                   |— IceSpell.java
```





## ## How To Compile

---

1. Open PowerShell and change into the project root: `cd LegendsOfValors-.../LegendsOfValors-....`
2. Create the compilation output directory if it does not exist: `mkdir out`.
3. Generate a response file listing every source: `Get-ChildItem -Path .\src -Recurse -Filter *.java | ForEach-Object { $_.FullName } | Set-Content -Path .\out\sources.txt`.
4. Compile all sources with UTF-8 encoding: `javac -encoding UTF-8 -d out -cp src @out\sources.txt`.
5. Launch the game: `java -cp out Main`.

## ## Notes

---

### ### Monster AI

- Each monster evaluates nearby hero positions every turn using risk-adjusted Manhattan distances.
- Decision types include advance, retreat, evade, and attack, prioritising survival when health or defense is low.
- Target selection favors wounded heroes first, then the highest offensive threat.
- Movement scoring balances lane progress, combat opportunities, and obstacle penalties before issuing a command.

## ## Input/Output Example

---

### ### Startup Menu and Mode Selection



### ### Legends of Valor mission briefing and map

# LEGENDS OF VALOR

## Map Legend

|        |                                         |
|--------|-----------------------------------------|
| Nexus  | Monster Nexus - Spawn point for enemies |
| Top    | Top Lane - Hero control zone            |
| Mid    | Mid Lane - Hero control zone            |
| Bot    | Bot Lane - Hero control zone            |
| Bush   | Bush - Increases Dexterity              |
| Cave   | Cave - Increases Agility                |
| Koulou | Koulou - Increases Strength             |
| Block  | Obstacle - Cleared when hero enters     |
| Wall   | Wall - Inaccessible                     |

H# = Hero Marker    M# = Monster Marker

### --- MISSION ---

1. Move your heroes to the Monster Nexus (Row 0) to win.
2. Defeat monsters to gain XP and Gold.
3. Buy better gear at your Nexus (Row 7).
4. Do not let monsters reach your Nexus!

Press ENTER to start the battle...

Press Enter to continue...

□

### ### Hero assignment, equipment tips, and turn interface

```
Select Hero #2 (Mid Lane):
ID   Name                Type      Lvl
-----
1    Gaerdal_Ironhand     [ALREADY SELECTED]
2    Sehanine_Monnbow     Warrior   1
3    Muamman_Duathall     Warrior   1
4    Flandal_Steelskin    Warrior   1
5    Undefeated_Yoj       Warrior   1
6    Eunoia_Cyn           Warrior   1
7    Rillifane_Rallathil  Sorcerer  1
8    Segojan_Earthcaller  Sorcerer  1
9    Reign_Havoc          Sorcerer  1
10   Reverie_Ashels       Sorcerer  1
11   Kalabar              Sorcerer  1
12   Skye_Soar            Sorcerer  1
13   Parzival             Paladin   1
14   Sehanine_Moonbow     Paladin   1
15   Skoraeus_Stonebones  Paladin   1
16   Garl_Glittergold     Paladin   1
17   Amaryllis_Astra      Paladin   1
18   Caliber_Heist        Paladin   1
Enter ID: 6
```

```
Select Hero #3 (Bot Lane):
ID   Name                Type      Lvl
-----
1    Gaerdal_Ironhand     [ALREADY SELECTED]
2    Sehanine_Monnbow     Warrior   1
3    Muamman_Duathall     Warrior   1
4    Flandal_Steelskin    Warrior   1
5    Undefeated_Yoj       Warrior   1
6    Eunoia_Cyn           [ALREADY SELECTED]
7    Rillifane_Rallathil  Sorcerer  1
8    Segojan_Earthcaller  Sorcerer  1
9    Reign_Havoc          Sorcerer  1
10   Reverie_Ashels       Sorcerer  1
11   Kalabar              Sorcerer  1
12   Skye_Soar            Sorcerer  1
13   Parzival             Paladin   1
14   Sehanine_Moonbow     Paladin   1
15   Skoraeus_Stonebones  Paladin   1
16   Garl_Glittergold     Paladin   1
17   Amaryllis_Astra      Paladin   1
18   Caliber_Heist        Paladin   1
Enter ID: 7
```

Here are some complimentary weapons for your heroes!

Gaerdal\_Ironhand - Equipped Weapon: Stick  
Eunoia\_Cyn - Equipped Weapon: Knife  
Rillifane\_Rallathil - Equipped Weapon: Melee

But remember, always head to market to get more skilled weapons!

-----  
Gaerdal\_Ironhand entered the Top Lane.  
Eunoia\_Cyn entered the Mid Lane.  
Rillifane\_Rallathil entered the Bot Lane.

Press Enter to continue...

☐

=====

ROUND 1

Next Monster Wave in: 7 rounds

=====

--- HERO SQUAD ---

| Name                | Lane | Pos | HP / MP    | Lvl | Gold |
|---------------------|------|-----|------------|-----|------|
| Gaerdal_Ironhand    | Top  | 7,0 | 100 / 100  | 1   | 1354 |
| Eunoia_Cyn          | Mid  | 7,3 | 100 / 400  | 1   | 2500 |
| Rillifane_Rallathil | Bot  | 7,6 | 100 / 1300 | 1   | 2500 |

--- ENEMY THREATS ---

| Lane | Closest Enemy | HP  | Dist |
|------|---------------|-----|------|
| Top  | Desghidorrah  | 100 | 7    |
| Mid  | Chrysophylax  | 100 | 7    |
| Bot  | BunsenBurner  | 100 | 7    |

-----

|    |  |  |    |  |  |    |  |
|----|--|--|----|--|--|----|--|
| M1 |  |  | M2 |  |  | M3 |  |
|    |  |  |    |  |  |    |  |
|    |  |  |    |  |  |    |  |
|    |  |  |    |  |  |    |  |
|    |  |  |    |  |  |    |  |
|    |  |  |    |  |  |    |  |
|    |  |  |    |  |  |    |  |
|    |  |  |    |  |  |    |  |
| H1 |  |  | H2 |  |  | H3 |  |

[H1] Action for Gaerdal\_Ironhand (Top Lane):

[W/A/S/D] Move | [T] Teleport | [K] Attack | [C] Cast Spell | [R] Recall

[M] Market | [I] Info | [E] Equip/Item | [L] Legend | [Q] Quit

> |



[H1] Action for Gaerdal\_Ironhand (Top Lane):

[W/A/S/D] Move | [T] Teleport | [K] Attack | [C] Cast Spell | [R] Recall

[M] Market | [I] Info | [E] Equip/Item | [L] Legend | [Q] Quit

> w

Moved to a CAVE tile : Gaerdal\_Ironhand received 10.0% increase in Agility

```
+-----+-----+-----+-----+-----+-----+-----+-----+
|M1|  |  |  |M2|  |  |  |M3|  |  |
+-----+-----+-----+-----+-----+-----+-----+-----+
|  |  |  |  |  |  |  |  |  |  |
+-----+-----+-----+-----+-----+-----+-----+-----+
|  |  |  |  |  |  |  |  |  |  |
+-----+-----+-----+-----+-----+-----+-----+-----+
|  |  |  |  |  |  |  |  |  |  |
+-----+-----+-----+-----+-----+-----+-----+-----+
|  |  |  |  |  |  |  |  |  |  |
+-----+-----+-----+-----+-----+-----+-----+-----+
|  |  |  |  |  |  |  |  |  |  |
+-----+-----+-----+-----+-----+-----+-----+-----+
|H1|  |  |  |  |  |  |  |  |  |
+-----+-----+-----+-----+-----+-----+-----+-----+
|  |  |  |  |  |  |  |  |  |  |
+-----+-----+-----+-----+-----+-----+-----+-----+
```

[H2] Action for Eunoia\_Cyn (Mid Lane):

[W/A/S/D] Move | [T] Teleport | [K] Attack | [C] Cast Spell | [R] Recall

[M] Market | [I] Info | [E] Equip/Item | [L] Legend | [Q] Quit

> w

Eunoia\_Cyn clears the obstacle! (Turn consumed)

```
+-----+-----+-----+-----+-----+-----+-----+-----+
|M1|  |  |  |M2|  |  |  |M3|  |  |
+-----+-----+-----+-----+-----+-----+-----+-----+
|  |  |  |  |  |  |  |  |  |  |
+-----+-----+-----+-----+-----+-----+-----+-----+
|  |  |  |  |  |  |  |  |  |  |
+-----+-----+-----+-----+-----+-----+-----+-----+
|  |  |  |  |  |  |  |  |  |  |
+-----+-----+-----+-----+-----+-----+-----+-----+
|  |  |  |  |  |  |  |  |  |  |
+-----+-----+-----+-----+-----+-----+-----+-----+
|  |  |  |  |  |  |  |  |  |  |
+-----+-----+-----+-----+-----+-----+-----+-----+
|H1|  |  |  |  |  |  |  |  |  |
+-----+-----+-----+-----+-----+-----+-----+-----+
|  |  |  |  |  |  |  |  |  |  |
+-----+-----+-----+-----+-----+-----+-----+-----+
```

[H3] Action for Rillifane\_Rallathil (Bot Lane):

[W/A/S/D] Move | [T] Teleport | [K] Attack | [C] Cast Spell | [R] Recall

[M] Market | [I] Info | [E] Equip/Item | [L] Legend | [Q] Quit

>

### ### Terrain Advantages, Obstacle Removal, and Combat Feedback

=== MONSTER TURN ===

None of the monsters are in range to attack.  
Monsters are advancing...

Press Enter to continue...

=== ROUND 1 ENDED ===

Heroes rested and recovered HP/Mana.

Press Enter to continue...

=====

ROUND 2

Next Monster Wave in: 6 rounds

=====

--- HERO SQUAD ---

| Name                | Lane | Pos | HP / MP    | Lvl | Gold |
|---------------------|------|-----|------------|-----|------|
| Gaerdal_Ironhand    | Top  | 6,0 | 111 / 100  | 1   | 1354 |
| Eunoia_Cyn          | Mid  | 7,3 | 111 / 400  | 1   | 2500 |
| Rillifane_Rallathil | Bot  | 6,6 | 111 / 1300 | 1   | 2500 |

--- ENEMY THREATS ---

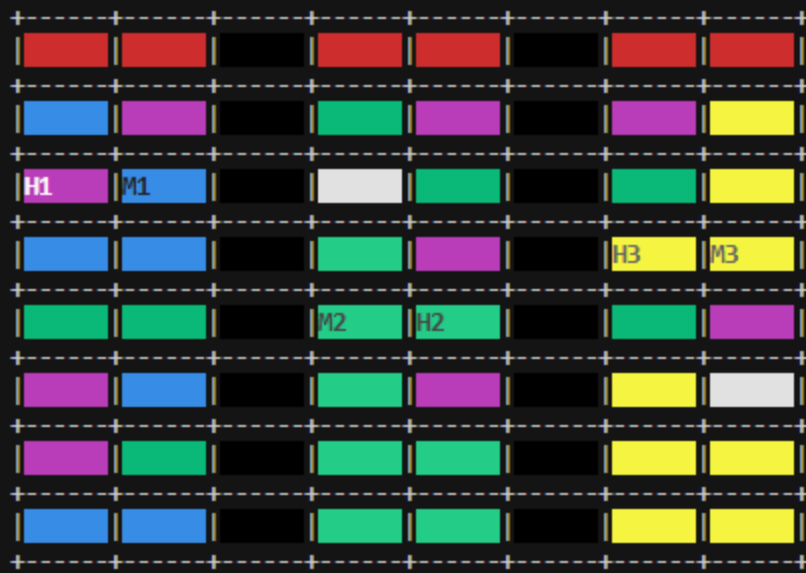
| Lane | Closest Enemy | HP  | Dist |
|------|---------------|-----|------|
| Top  | Desghidorrah  | 100 | 6    |
| Mid  | Chrysophylax  | 100 | 6    |
| Bot  | BunsenBurner  | 100 | 6    |

```
+-----+-----+-----+-----+-----+-----+-----+
|  |  |  |  |  |  |  |  |
+-----+-----+-----+-----+-----+-----+-----+
|M1 |  |  |  | M2 |  |  | M3 |  |
+-----+-----+-----+-----+-----+-----+-----+
|  |  |  |  |  |  |  |  |
+-----+-----+-----+-----+-----+-----+-----+
|  |  |  |  |  |  |  |  |
+-----+-----+-----+-----+-----+-----+-----+
|  |  |  |  |  |  |  |  |
+-----+-----+-----+-----+-----+-----+-----+
|  |  |  |  |  |  |  |  |
+-----+-----+-----+-----+-----+-----+-----+
|  |  |  |  |  |  |  |  |
+-----+-----+-----+-----+-----+-----+-----+
|H1 |  |  |  |  |  |  | H3 |  |
+-----+-----+-----+-----+-----+-----+-----+
|  |  |  |  |  |  |  |  |
+-----+-----+-----+-----+-----+-----+-----+
|  |  |  |  |  |  |  |  |
+-----+-----+-----+-----+-----+-----+-----+
```

[H1] Action for Gaerdal\_Ironhand (Top Lane):

[W/A/S/D] Move | [T] Teleport | [K] Attack | [C] Cast Spell | [R] Recall  
[M] Market | [I] Info | [E] Equip/Item | [L] Legend | [Q] Quit

>



```

=== MONSTER TURN ===
?? Desghidorrah attacks Gaerdal_Ironhand!

[ ATTACK ]
Desghidorrah hit Gaerdal_Ironhand for 100 damages.

?? Chrysophylax attacks Eunoia_Cyn!

[ ATTACK ]
Chrysophylax hit Eunoia_Cyn for 100 damages.

?? BunsenBurner attacks Rillifane_Rallathil!

[ ATTACK ]
BunsenBurner hit Rillifane_Rallathil for 100 damages.

Monsters are advancing...

Press Enter to continue...

```

[H1] Action for Gaerdal\_Ironhand (Top Lane):

[W/A/S/D] Move | [T] Teleport | [K] Attack | [C] Cast Spell | [R] Recall  
[M] Market | [I] Info | [E] Equip/Item | [L] Legend | [Q] Quit

> k

?? Gaerdal\_Ironhand attacks Desghidorrah!

[ ATTACK ]

1. Stick (Damage: 50)

Select Weapon (0 cancel): <> 1

Gaerdal\_Ironhand dealt 34 damages to Desghidorrah.

=== MONSTER TURN ===

?? Chrysophylax attacks Eunoia\_Cyn!

[ ATTACK ]

Chrysophylax attacked Gaerdal\_Ironhand -> BUT MISSED! Gaerdal\_Ironhand dodged the attack!

?? BunsenBurner attacks Rillifane\_Rallathil!

[ ATTACK ]

BunsenBurner hit Eunoia\_Cyn for 100 damages.

...

Rillifane\_Rallathil has fainted

...

BunsenBurner has defeated Rillifane\_Rallathil !

Eunoia\_Cyn has respawned at their home nexus.

### Teleportation Support and High Turn Information Display

[H1] Action for Gaerdal\_Ironhand (Top Lane):

[W/A/S/D] Move | [T] Teleport | [K] Attack | [C] Cast Spell | [R] Recall  
[M] Market | [I] Info | [E] Equip/Item | [L] Legend | [Q] Quit

> t

Select a hero to teleport to:

1. Eunoia\_Cyn (Lane: Mid)

2. Rillifane\_Rallathil (Lane: Bot)

Enter Target ID (0 to cancel): 1

Teleporting Gaerdal\_Ironhand to (7, 4)...

|    |  |  |    |    |  |    |    |
|----|--|--|----|----|--|----|----|
|    |  |  |    |    |  |    |    |
| M1 |  |  |    |    |  |    |    |
|    |  |  |    |    |  |    | M3 |
|    |  |  |    |    |  | H3 |    |
|    |  |  |    |    |  |    |    |
|    |  |  |    |    |  |    |    |
|    |  |  | M2 |    |  |    |    |
|    |  |  |    |    |  |    |    |
|    |  |  |    |    |  |    |    |
|    |  |  |    |    |  |    |    |
|    |  |  | H2 | H1 |  |    |    |

[H2] Action for Eunoia\_Cyn (Mid Lane):

[W/A/S/D] Move | [T] Teleport | [K] Attack | [C] Cast Spell | [R] Recall  
[M] Market | [I] Info | [E] Equip/Item | [L] Legend | [Q] Quit

>

=====

ROUND 14

Next Monster Wave in: 2 rounds

=====

--- HERO SQUAD ---

| Name                | Lane | Pos | HP / MP    | Lvl | Gold |
|---------------------|------|-----|------------|-----|------|
| Gaerdal_Ironhand    | Top  | 5,0 | 134 / 100  | 1   | 1354 |
| Eunoia_Cyn          | Mid  | 6,3 | 122 / 400  | 1   | 2500 |
| Rillifane_Rallathil | Bot  | 4,6 | 147 / 1300 | 1   | 2500 |

--- ENEMY THREATS ---

| Lane | Closest Enemy | HP  | Dist |
|------|---------------|-----|------|
| Top  | Desghidorrah  | 66  | 4    |
| Mid  | Chrysophylax  | 100 | 4    |
| Bot  | BunsenBurner  | 100 | 3    |

-----

|    |    |  |    |    |  |    |    |
|----|----|--|----|----|--|----|----|
|    |    |  |    |    |  |    |    |
|    |    |  |    |    |  |    |    |
|    |    |  |    |    |  |    |    |
|    |    |  |    |    |  |    |    |
| M4 | M1 |  | M2 | M5 |  |    | M6 |
|    |    |  |    |    |  | H3 | M3 |
| H1 |    |  |    |    |  |    |    |
|    |    |  | H2 |    |  |    |    |
|    |    |  |    |    |  |    |    |
|    |    |  |    |    |  |    |    |

[H1] Action for Gaerdal\_Ironhand (Top Lane):

[W/A/S/D] Move | [T] Teleport | [K] Attack | [C] Cast Spell | [R] Recall  
[M] Market | [I] Info | [E] Equip/Item | [L] Legend | [Q] Quit

>

□

####victory (Heroes win)

A 10x8 grid of colored squares representing a 1000x1000 matrix. The grid is divided into four 5x4 quadrants. The top-left quadrant (rows 1-5, columns 1-4) contains red, blue, and black squares. The top-right quadrant (rows 1-5, columns 5-8) contains red, black, and blue squares. The bottom-left quadrant (rows 6-10, columns 1-4) contains blue, black, and white squares. The bottom-right quadrant (rows 6-10, columns 5-8) contains black, white, and blue squares. The grid is labeled with 'H1' in the top-left, 'H2/M5' in the top-right, 'H3/M6' in the bottom-left, and 'H4' in the bottom-right.

[H1] Action for Sehanine\_Monnbow (Top Lane):

[W/A/S/D] Move | [T] Teleport | [K] Attack | [C] Cast Spell | [R] Recall

[M] Market | [I] Info | [E] Equip/Item | [L] Legend | [Q] Quit

 $\lambda > W$ 

A 10x8 grid of color patches for color calibration. The patches are arranged in rows and columns, with some patches labeled H1, H2/M5, and H3/M6. The colors include red, blue, green, yellow, black, white, and purple.

```

=====
[Victory] Sehanine Monnbow reached the Nexus!
=====

The heroes have won the game against the monsters.
+ NUMBER OF ROUNDS PLAYED: 14
Hero Stats:
Sehanine_Monnbow | HP: 429.0 | Level: 1 | Lane: Top | Nexus: (7, 0)
Flandar_Steelskin | HP: 323.0 | Level: 1 | Lane: Mid | Nexus: (7, 3)
Undefeated_Yoj | HP: 65.0 | Level: 1 | Lane: Bot | Nexus: (7, 6)

NUMBER OF MONSTERS DEFEATED: 7
-----
Do you want to play another game? [Y] [N]

```

###victory (Monsters win)

```

[ ATTACK ]
Chrysophylax attacked Undefeated_Yoj -> BUT MISSED! Undefeated_Yoj dodged the attack!
Monsters are advancing...

*****
GAME OVER! A Monster breached the Nexus!
*****

The heroes have lost the game against the monsters.
+ NUMBER OF ROUNDS PLAYED: 7
Hero Stats:
Flandar_Steelskin | HP: 111.0 | Level: 1 | Lane: Top | Nexus: (7, 0)
Undefeated_Yoj | HP: 45.0 | Level: 1 | Lane: Mid | Nexus: (7, 3)
Eunoia_Cyn | HP: 178.0 | Level: 1 | Lane: Bot | Nexus: (7, 6)

NUMBER OF MONSTERS DEFEATED: 3
-----
Do you want to play another game? [Y] [N]

```